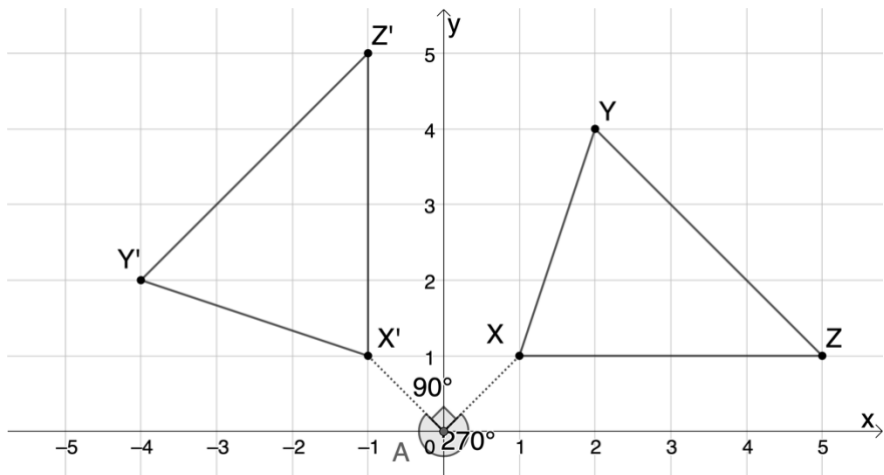


On the coordinate plane shown, $\triangle XYZ$ is rotated to create $\triangle X'Y'Z'$. Use this image for questions 1-6.



There are two ways to describe the rotation. Complete both statements using the information in the given figure.

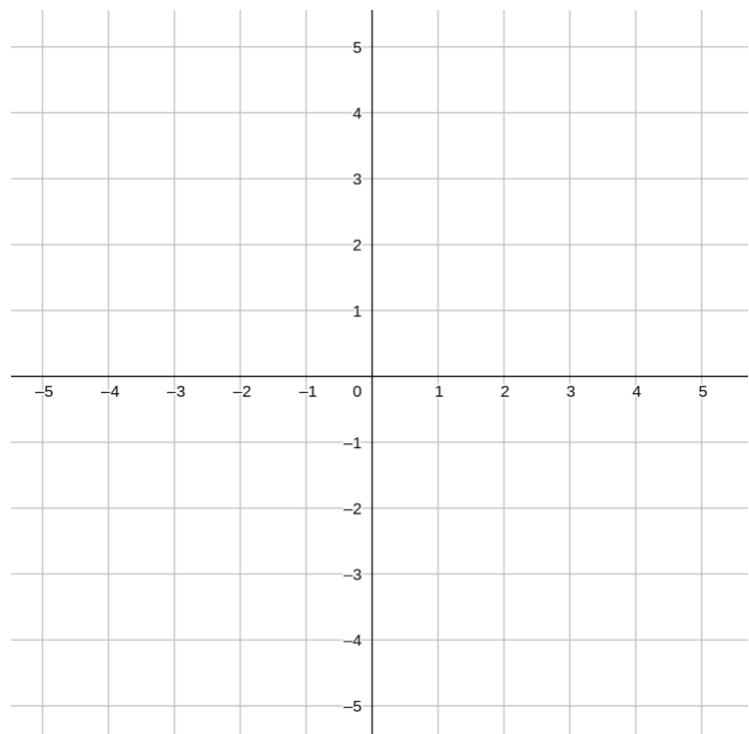
1. $\triangle XYZ$ is rotated _____ $^\circ$ clockwise about the origin to create $\triangle X'Y'Z'$.
2. $\triangle XYZ$ is rotated _____ $^\circ$ counterclockwise about the origin to create $\triangle X'Y'Z'$.

Complete the table with the coordinates of each vertex.

Coordinates		Coordinates	
3. Point X		Point X'	
4. Point Y		Point Y'	
5. Point Z		Point Z'	

6. Describe the relationship between the coordinates of the preimage and the image.

7. The vertices of $\triangle ABC$ have their coordinates given in the below table. Plot $\triangle ABC$ on the coordinate grid.



Rotate $\triangle ABC$ 180° clockwise with the origin as the center of rotation. Label the image $\triangle A'B'C'$ and complete the table for the image.

Coordinates		Coordinates	
8. Point A	(−3,1)	Point A'	
9. Point B	(−1,2)	Point B'	
10. Point C	(−2,−2)	Point C'	